

ODD 2024

Tutorial Sheet - 1

Software Development Fundamentals – I (15B11CI111)

COURSE OUTCOMES		COGNITIVE LEVELS
CO1	Explain the logic for solving problems considering various phases of software development life cycle and depicting them using algorithms and flowcharts	Understand (Level 2)
CO2	Explain basics of C programming concepts to make decision for solving problems	Understand (Level 2)
CO3	Demonstrate and contrast different methods for writing modular programs in C	Understand (Level 2)
CO4	Use various C programming constructs to implement iteration, and recursion	Apply (Level 3)
CO5	Apply and implement arrays, pointers, structures and file handling for solving real-world problems	Apply (Level 3)

Note: Students are advised to submit their solutions to their respective tutorial faculties.

Q1. [CO1] Read the given information carefully and answer the questions that follow. Eleven friends M, N, O, P, Q, R, S, T, U, V and W are sitting in the first row of the stadium watching a cricket match.

T is to the immediate left of P and third to the right of U.

V is the immediate neighbour of M and N and third to the left of S.

M is the second to the right of Q, who is at one of the ends.

R is sitting next to the right of P and P is second to the right of O.

(a) Who is sitting in the center of the row?

1. N 2. O 3. S 4. U

(b) Which of the following people are sitting to the right of S?

1. OTPQ 2. OTPR 3. UNVM 4. UOTPR

(c) Which of the following statements is true with respect to the above arrangement?

1. There are three persons sitting between P and S
2. W is between M and V.
3. N is sitting between V and U
4. S and O are neighbours sitting to the immediate right of T

(d) Who are the immediate neighbours of T?

1. O, P 2. O, R 3. N, U 4. V, U

(e) If Q and P, O and N, M and T, and W and R interchange their positions then which of the following pairs of friends is sitting at the ends?

1. P and Q 2. Q and R 3. P and W 4. W and R

Q2. [CO1] A man has 53 socks in his drawer: 21 identical blue, 15 identical black and 17 identical red. The lights are out and he is completely in the dark. How many socks must he take out to make 100 percent certain he has at least one pair of black socks?

Q3. [CO1] If A is substituted by 4, B by 3, C by 2, D by 4, E by 3, F by 2 and so on, then what will be total of the numerical values of the letters of the word SICK?

Analyse the Requirements and Flow of Events for Q4 & Q5

Q4. [CO1] Lift / Elevator control system in multi-story building.

Q5. [CO1] Intelligent traffic control signal system to control the traffic movements at squares/junction.